Adaptive LiDAR terrain mapping

Proposed FPGA architecture for adaptive 2.5D elevation maps

MICROMI

Host workstation

Recorded LiDAR frames
Retain original point records
Replay one frame in two passes

Gigabit Ethernet

Semantic preprocessing
Optional class and confidence
CNN inference on the host

Geometry-based operation
Available without semantic labels

Map evaluation
Elevation map and validity mask
Compare software reference maps
Measure time and allocated bytes

RISC-V processor subsystem

Frame receiver
Verify frame ID and point count
Store point records in shared memory

Transfer and replay control
Configure buffers, schedule passes, export maps

Readback

Pass configuration

Shared LPDDR4 memory

Replay buffer
Original point records
Aligned labels when supplied

Map storage
Coarse cells and fine tiles
Fixed-capacity allocation table

Export buffer
Elevation map and status
Frame counters

MSS memory controller arbitrates processor and fabric access.
Software assigns buffer ownership before each transfer.

Read retained points and cell state

Write map state and counters

FPGA terrain-mapping engine

AXI interface and transfer engine
Read retained frames and cell records. Write completed maps and status.

First pass builds coarse cells and selects fine tiles.
Second pass replays the original points into the selected cells.

Tile plan and map

Point validation
Reject invalid coordinates
Crop a 32 m square

Coordinate binning
Fixed-point cell addresses
Selected tile lookup

Cell accumulation
Minimum and maximum height
Height sum and point count
Per-cell observation flags

Refinement analysis
Height range and coverage
Optional host label cues

Tile allocation
Use a fixed-capacity tile pool
Keep coarse cells at capacity

Mapping region
32 m by 32 m

Initial cell sizes
10 cm, 20 cm and 40 cm

Refinement extension
5 cm cells